UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

ECE408 - CS483 - CSE408 Spring 2026
Applied Parallel Programming

Lecture 20: Data Transfer and CUDA Streams (Task Parallelism)

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Course Reminders

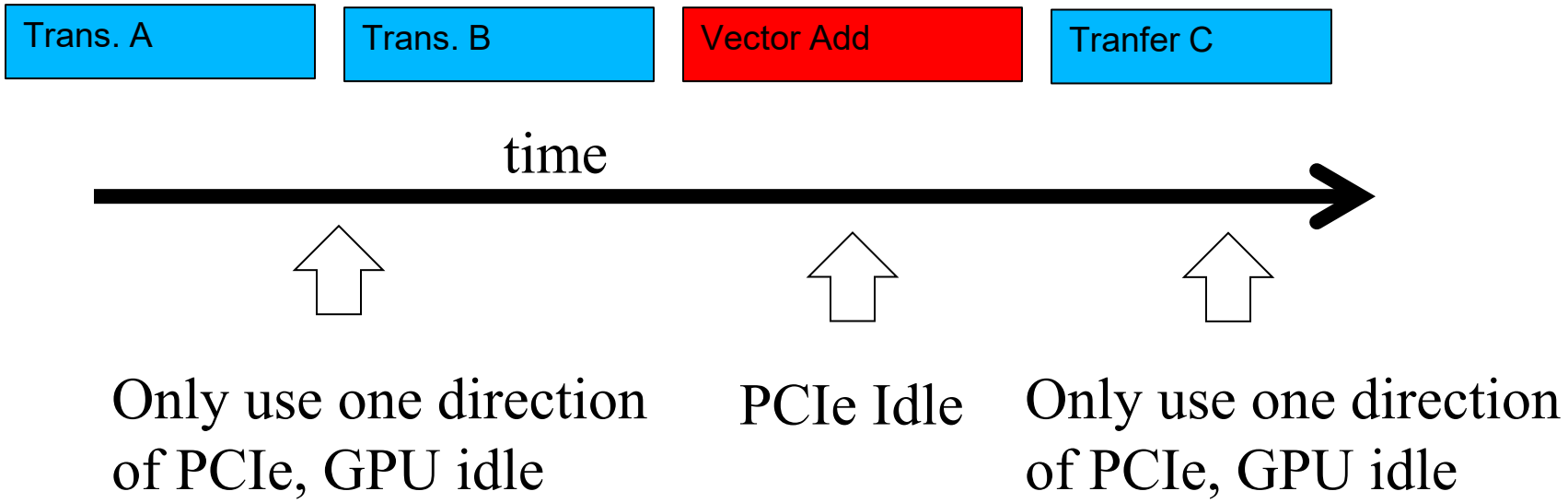
- Project
  - Project milestone 2 is due this Friday
- Labs
  - Lab 8 (last lab) will be due next week
- Midterm 2
  - 5/5/26

Today's Objectives

- To learn more advanced features of the CUDA APIs for data transfer and kernel launch
  - Task parallelism for overlapping data transfer with kernel computation
  - CUDA streams

Serialized Data Transfer

- So far, the way we use `cudaMemcpy` serializes data transfer and GPU computation



Device Overlap

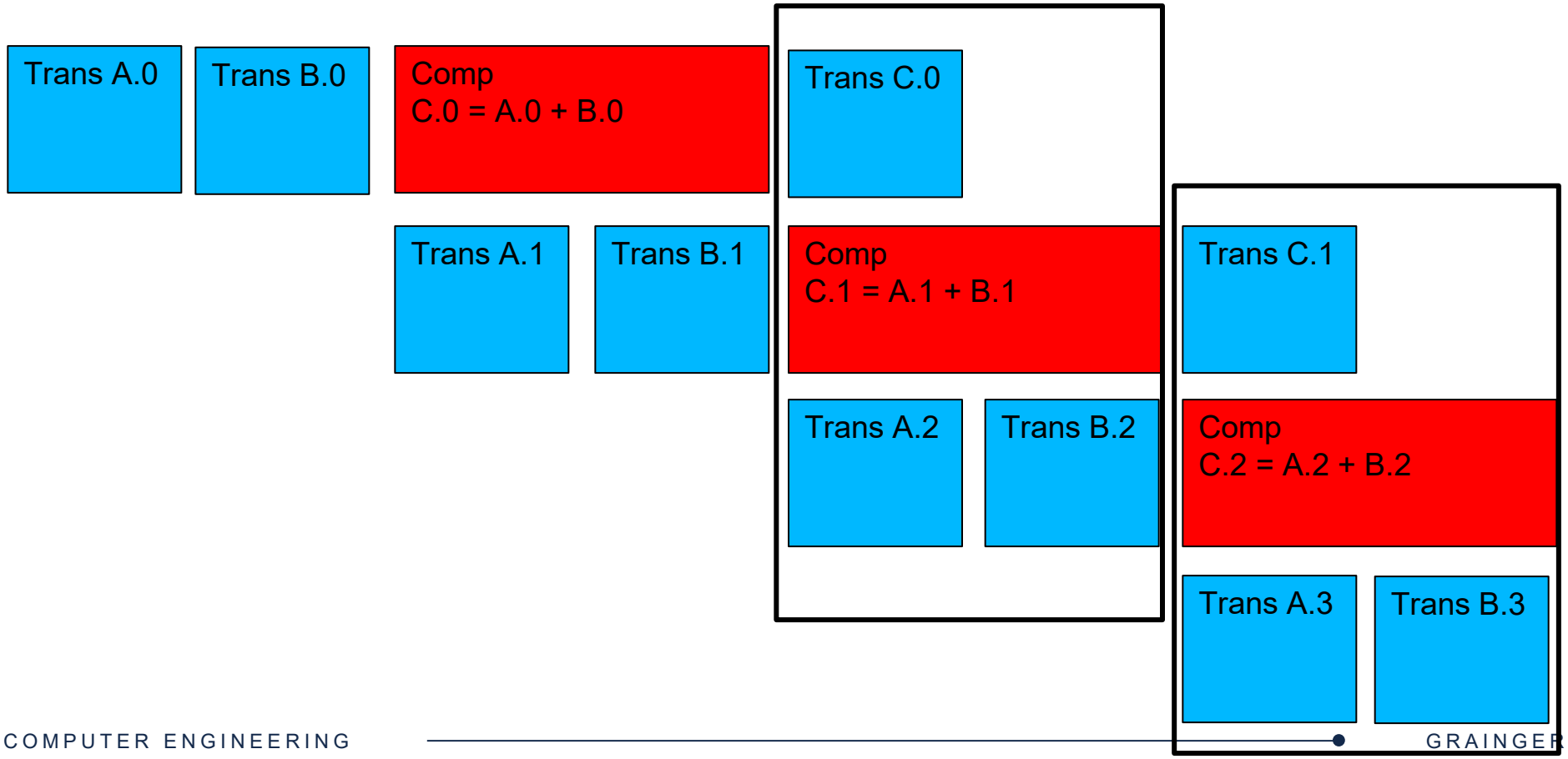
- Most CUDA devices support *device overlap*
  - Simultaneously execute a kernel while performing a copy between device and host memory

- Checking if this is supported

```
int dev_count;
cudaDeviceProp prop;

cudaGetDeviceCount( &dev_count);
for (int i = 0; i < dev_count; i++) {
  cudaGetDeviceProperties(&prop, i);

  if (prop.deviceOverlap) …
```

Overlapped (Pipelined) Timing

- Divide large vectors into segments
- Overlap transfer and compute of adjacent segments

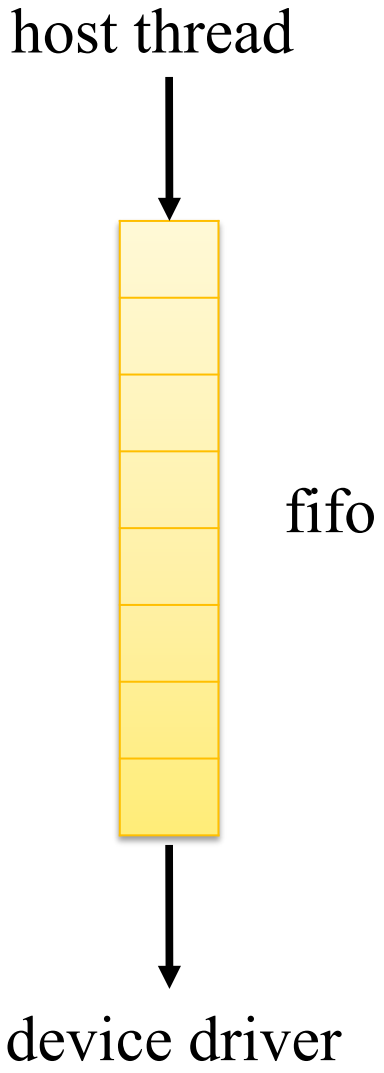


Using CUDA Streams and Asynchronous Memcpy

- CUDA supports parallel execution of kernels and cudaMemcpy with **streams**
- Each stream **is a queue of operations** (kernel launches and cudaMemcpy's)
- Operations (tasks) in different streams
  - can execute in parallel
  - a version of **task parallelism**

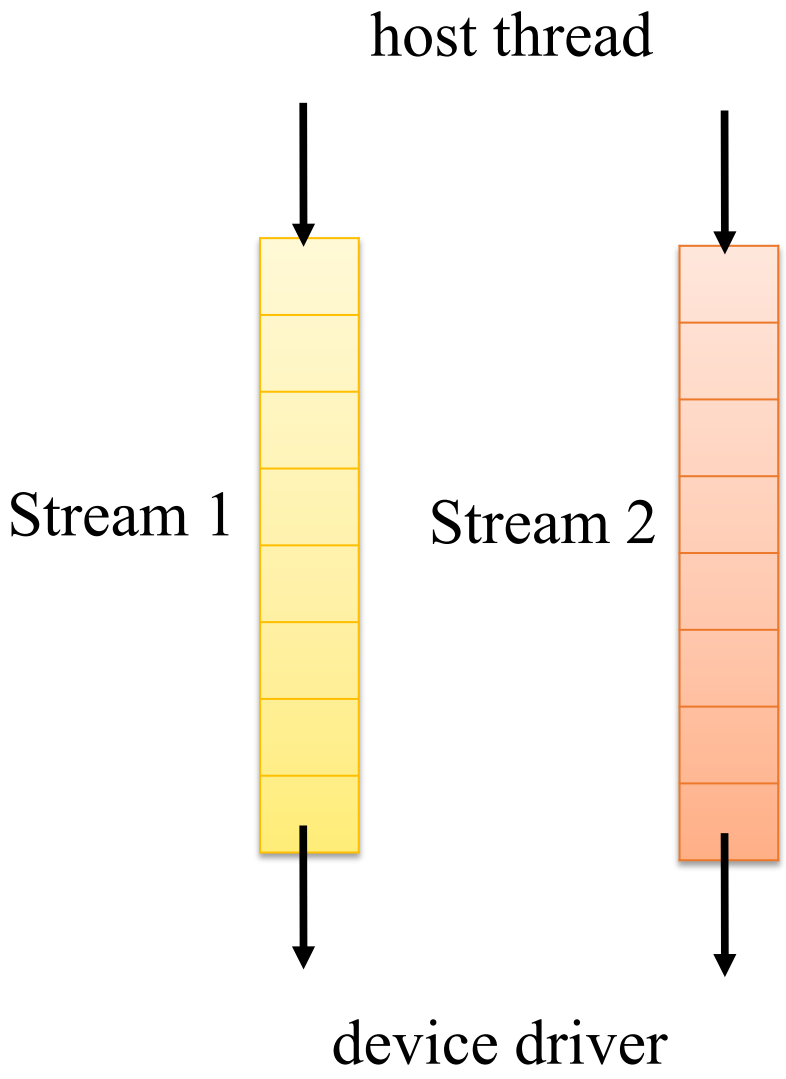
Streams

- Device requests made from the host code are put into a queue
  - Queue processed asynchronously by the driver and device. Called a “Stream”.
  - Driver ensures that commands in the queue are processed strictly in sequence. Memory copies end before kernel launch, etc.

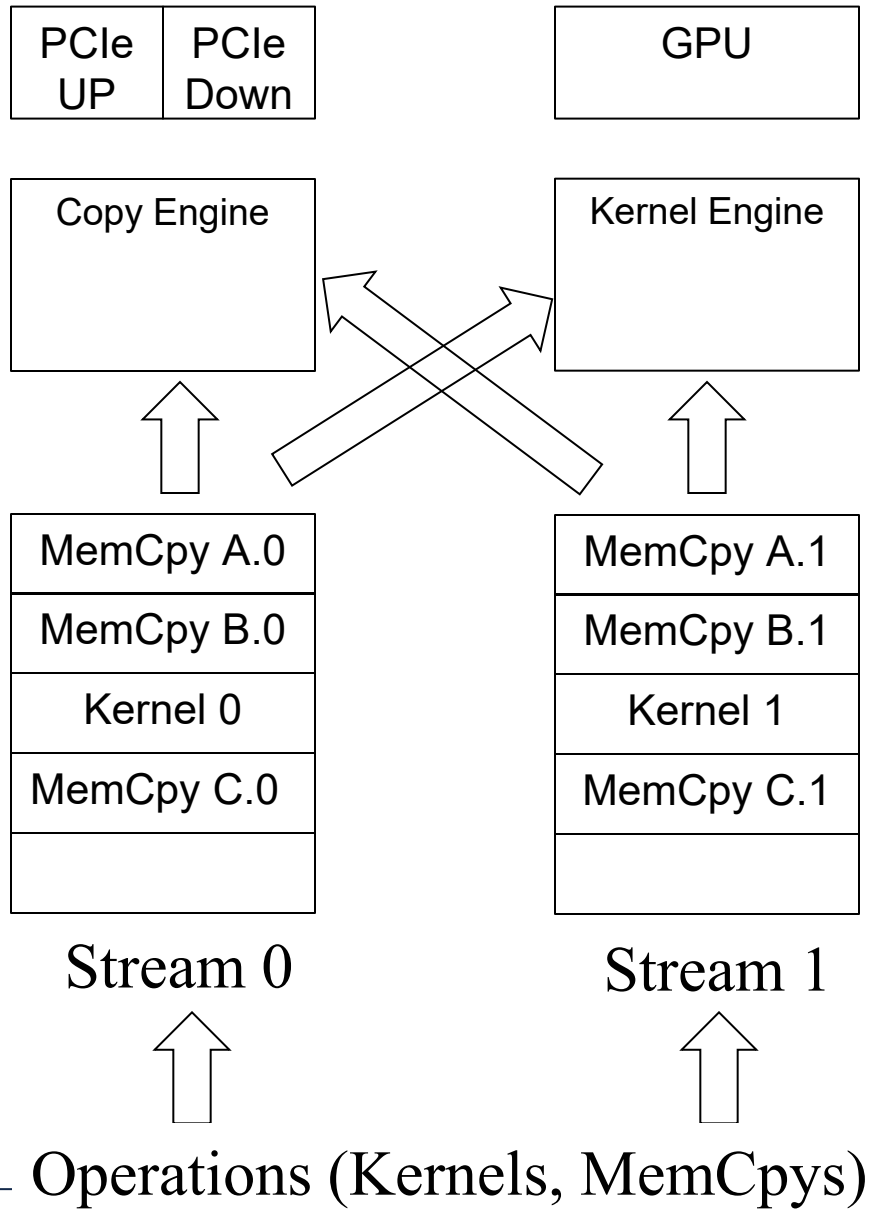


Multiple Streams

- To allow concurrent copying and kernel execution, multiple queues are required



Conceptual View of Streams



A Simple Multi-Stream Host Code

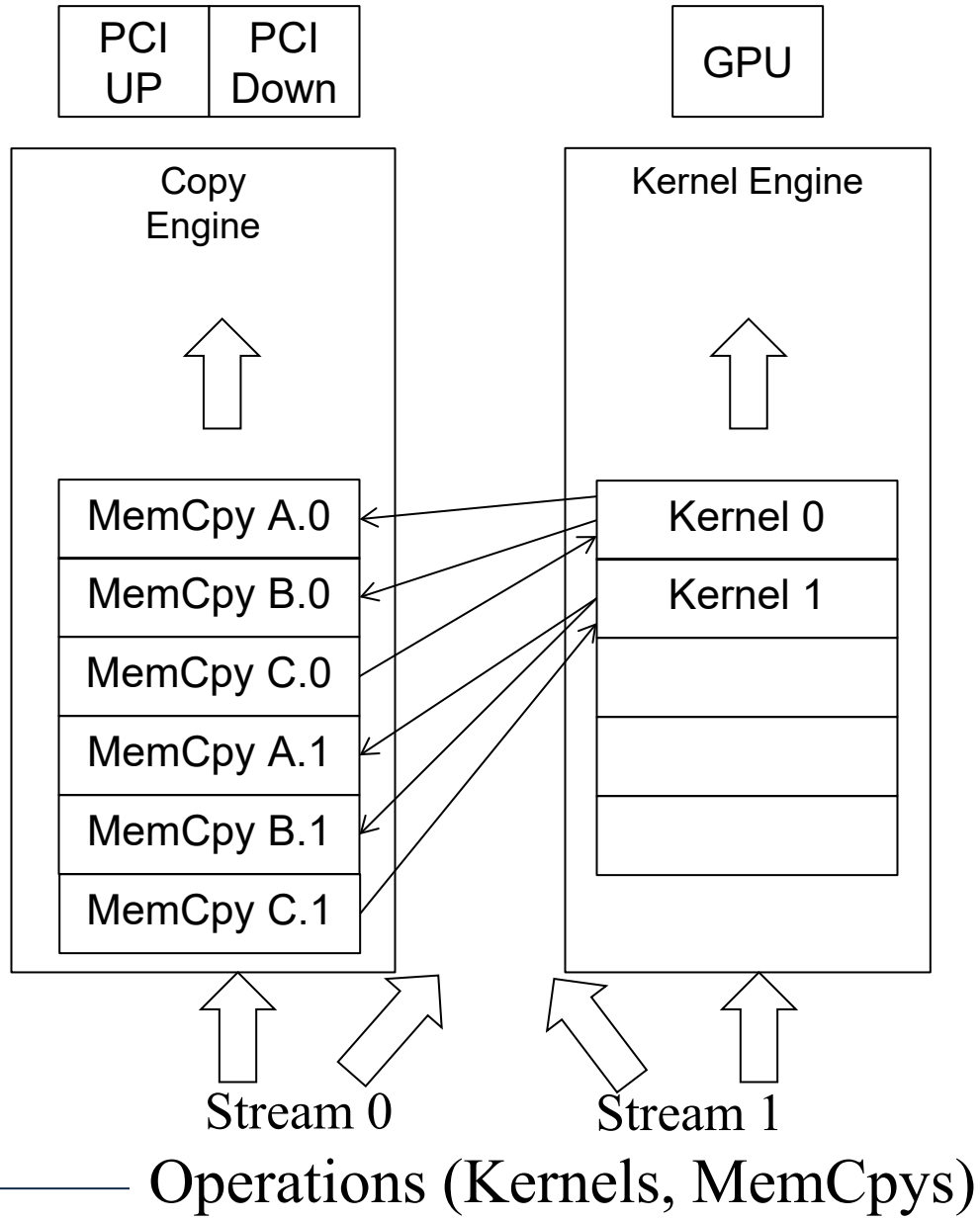
```
cudaStream_t stream0, stream1;
cudaStreamCreate(&stream0);
cudaStreamCreate(&stream1);
float *d_A0, *d_B0, *d_C0;     // device memory for stream 0
float *d_A1, *d_B1, *d_C1;     // device memory for stream 1

// cudaMalloc for d_A0, d_B0, d_C0, d_A1, d_B1, d_C1 go here

for (int i=0; i<n; i+=SegSize*2) {
  // copy data in stream0
  // launch kernel in stream0
  // copy results in stream0
  // copy data in stream1
  // launch kernel in stream1
  // copy results in stream1
}
```

A Simple Multi-Stream Host Code

```
for (int i=0; i<n; i+=SegSize*2)
{
  cudaMemcpyAsync(d_A0, h_A+i, SegSize*sizeof(float),.., stream0);
  cudaMemcpyAsync(d_B0, h_B+i, SegSize*sizeof(float),.., stream0);

  vecAdd<<<SegSize/256, 256, 0, stream0>>>(d_A0, d_B0, …);
  cudaMemcpyAsync(h_C+i, d_C0, SegSize*sizeof(float),.., stream0);

  cudaMemcpyAsync(d_A1, h_A+i+SegSize,
                                SegSize*sizeof(float),.., stream1);
  cudaMemcpyAsync(d_B1, h_B+i+SegSize,
                                SegSize*sizeof(float),.., stream1);

  vecAdd<<<SegSize/256, 256, 0, stream1>>>(d_A1, d_B1, …);
  cudaMemcpyAsync(h_C+i+SegSize, d_C1,
                                SegSize*sizeof(float),.., stream1);
}
```

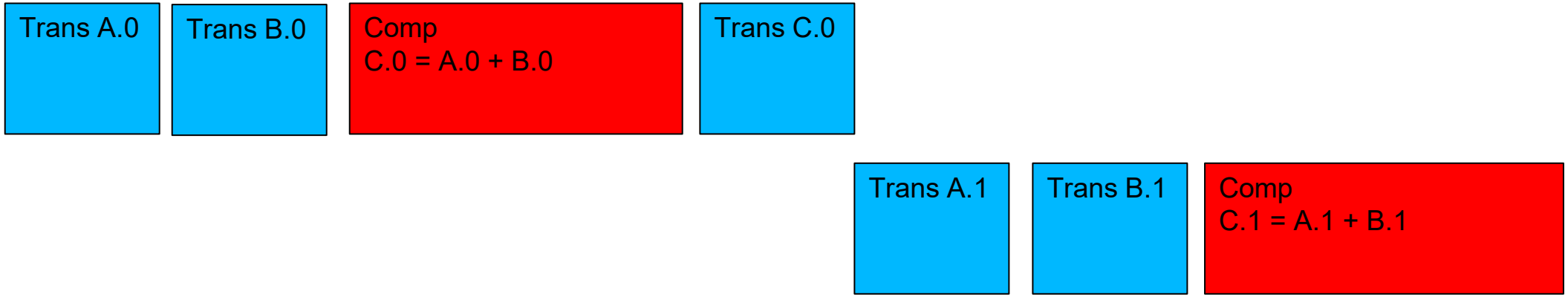
Older GPUs Support Streams in Software

- Task at head of queue waits for dependencies (arcs) before executing
  - For example, Kernel 0 waits for MemCpy A.0 and MemCpy B.0 to finish.



Not quite the overlap we want

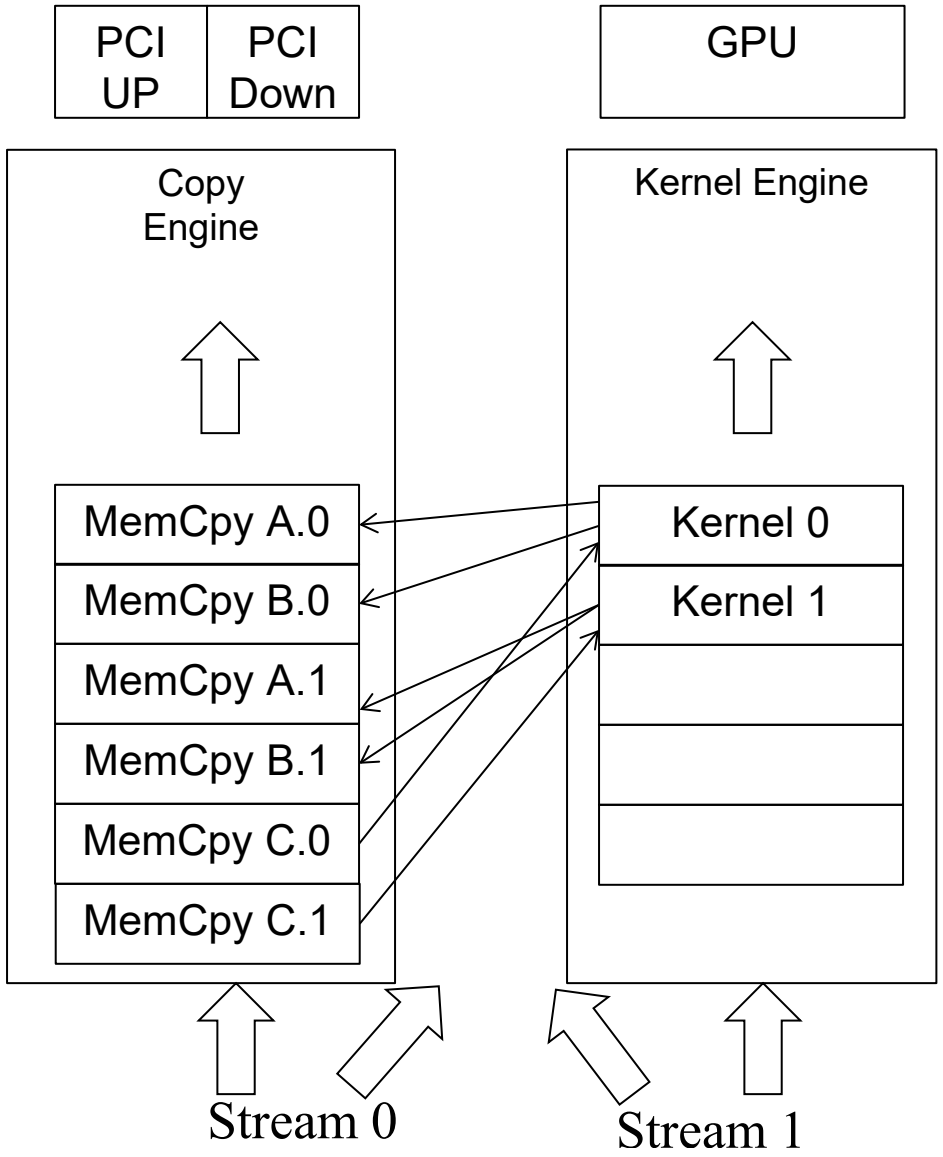
- C.0 blocks A.1 and B.1 in the copy engine queue



A Better Multi-Stream Host Code

```
for (int i=0; i<n; i+=SegSize*2) {

  cudaMemcpyAsync(d_A0, h_A+i; SegSize*sizeof(float),.., stream0);
  cudaMemcpyAsync(d_B0, h_B+i; SegSize*sizeof(float),.., stream0);
  cudaMemcpyAsync(d_A1, h_A+i+SegSize,
                              SegSize*sizeof(float),.., stream1);
  cudaMemcpyAsync(d_B1, h_B+i+SegSize,
                              SegSize*sizeof(float),.., stream1);

  vecAdd<<<SegSize/256, 256, 0, stream0>>>(d_A0, d_B0, …);
  vecAdd<<<SegSize/256, 256, 0, stream1>>>(d_A1, d_B1, …);

  cudaMemcpyAsync(d_C0, h_C+i; SegSize*sizeof(float),.., stream0);
  cudaMemcpyAsync(d_C1, h_C+i+SegSize,
                              SegSize*sizeof(float),.., stream1);
}
```

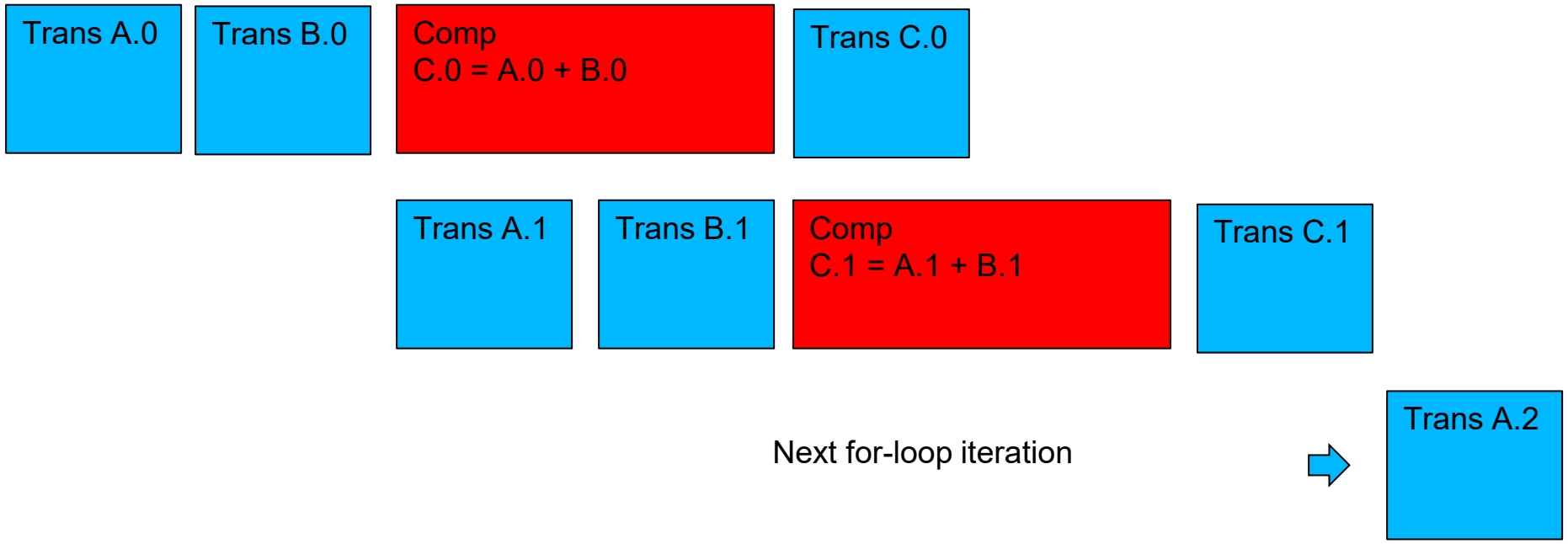
A View Closer to Reality



Operations (Kernels, MemCpys)

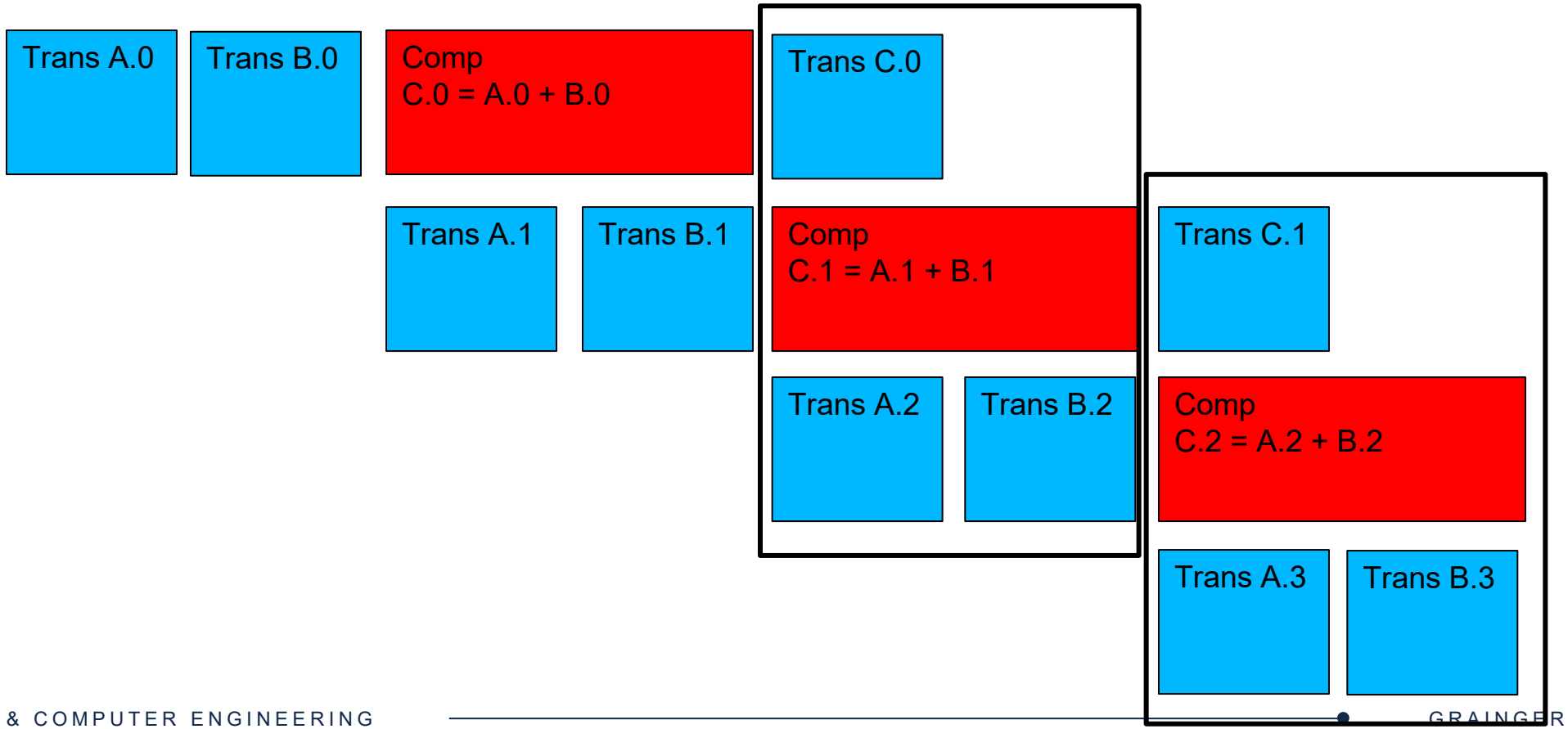
Better Overlap with Two Streams

- C.0 no longer blocks A.1 and B.1 in the copy engine queue
- However, C.1 still blocks A.2 and B.2 from the next iteration – PCIe used for only one direction



Three streams needed for continuous pipelining

- Divide large vectors into segments
- Overlap transfer and compute of adjacent segments

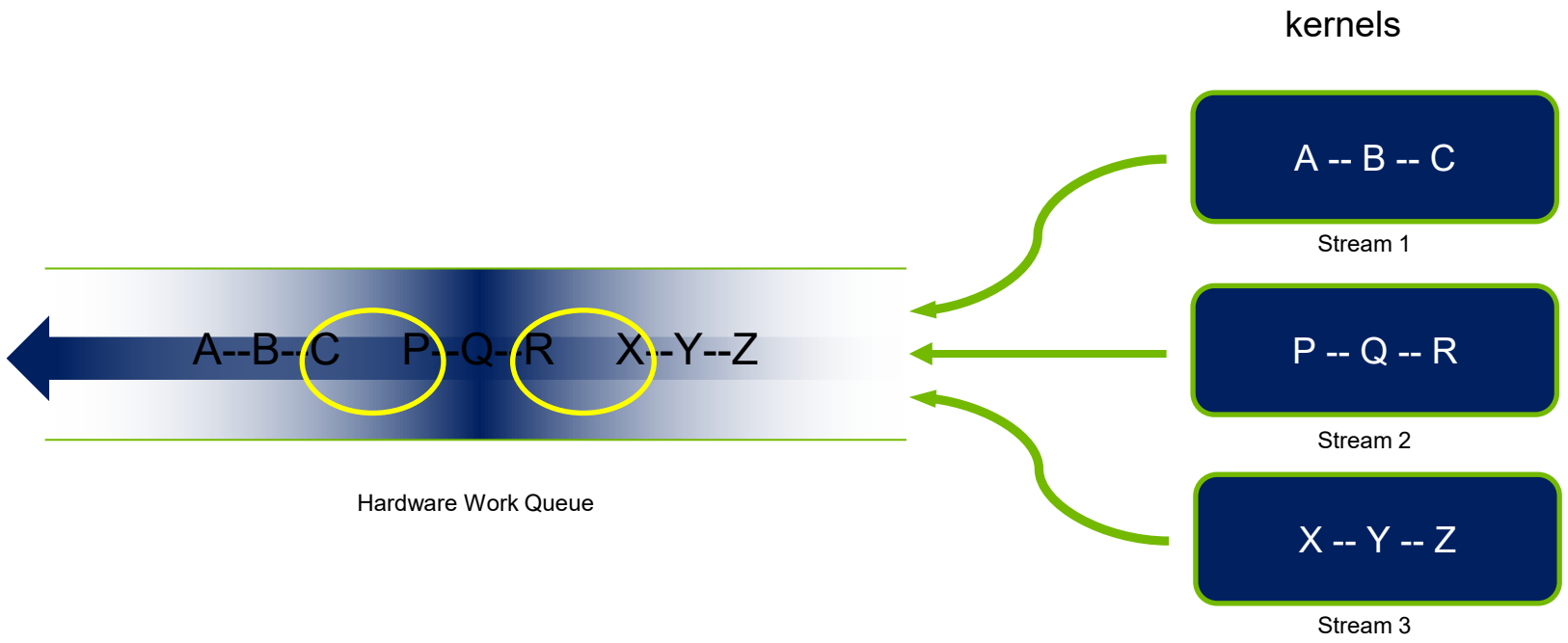


Hyper-Q

- Hyper-Q refers to a hardware feature introduced with the Kepler architecture (Compute Capability 3.5 and above) that significantly improves GPU utilization by enabling multiple CPU threads or processes to simultaneously submit work to the GPU.
  - Hyper-Q provides the hardware capability to execute work from multiple streams truly in parallel
  - Provides multiple real stream queues for each engine
  - Allows more concurrency by allowing some streams to make progress for an engine while others are blocked

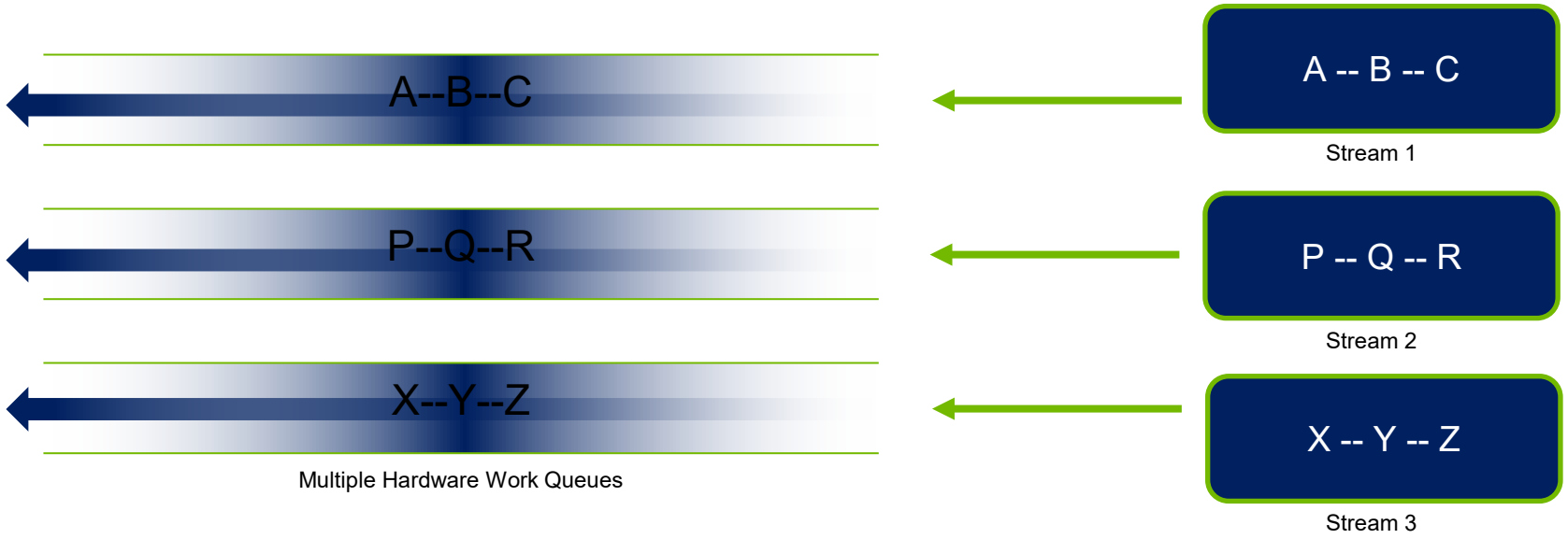
Fermi (and older) Concurrency

- Fermi allows 16-way concurrency
  - Up to 16 grids can run at once
  - But kernels from CUDA streams multiplex into a single queue
  - Overlap only at stream edges



Kepler Improved Concurrency

- Kepler allows 32-way concurrency
  - One work queue per stream
  - Concurrency at full-stream level
  - No inter-stream dependencies



Other considerations when working with streams

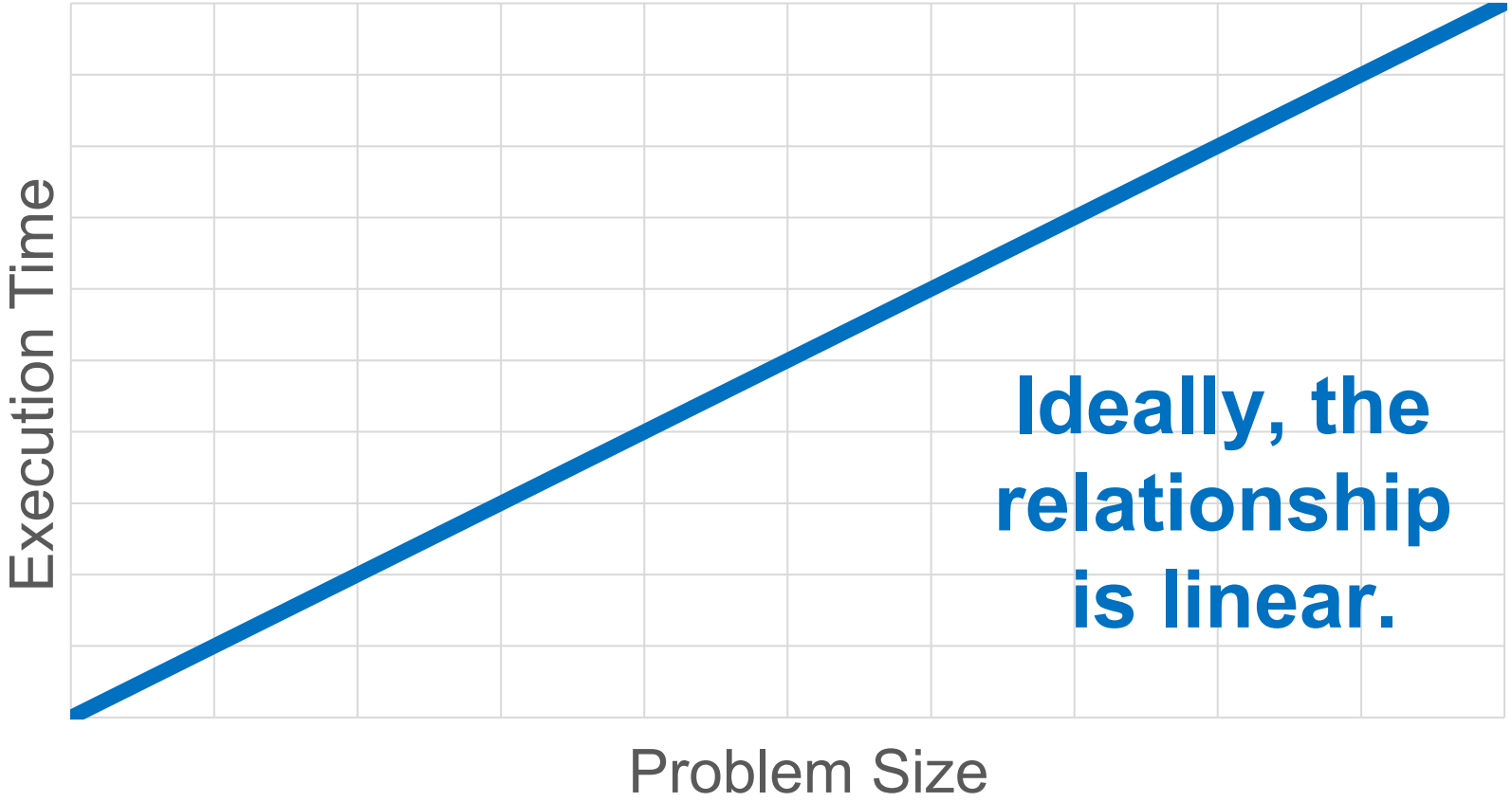
- cudaMemcpyAsync silently falls back to synchronous behavior if the host memory is not pinned
  - Use cudaMallocHost() / cudaFreeHost() instead of regular malloc() / free()
- Stream Synchronization Primitives
  - cudaStreamSynchronize(stream) — wait for a specific stream
  - cudaDeviceSynchronize() — wait for all streams
  - cudaStreamQuery(stream) — non-blocking poll
  - cudaEventRecord / cudaEventSynchronize — fine-grained event-based sync (this also opens the door to cross-stream dependencies, which is a natural follow-up)
- The Default (NULL) Stream
  - Has implicit synchronization behavior that can silently break your code
- Stream Destruction / Cleanup
  - Use cudaStreamDestroy() to destroy the streams once done

Smaller Segments Reduce Boundary Effects

- **How small should segments be?**
- If we **overlap**
  - **transfer of** segment N's **inputs**,
  - **computation** of segment N – 1, **and**
  - **transfer** of segment N – 2's **results**,
- we **still have non-overlapping work** at the beginning and the end.
- **So segments should be really small?**

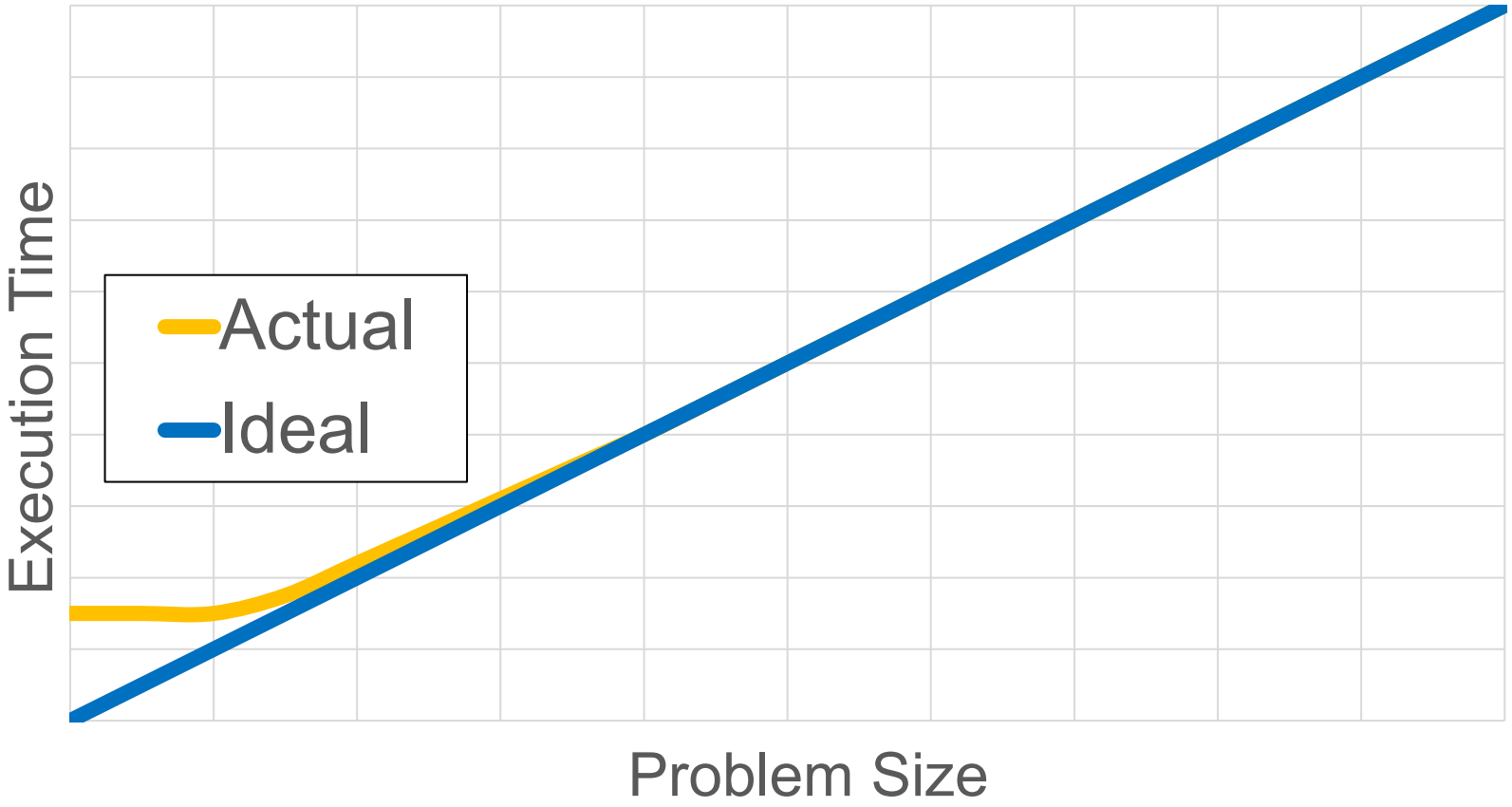
Execution Time is Ideally Linear in Size

- Think about execution time as a function of segment size.



Execution Time Never Reaches Zero

- But real execution time has a minimum.



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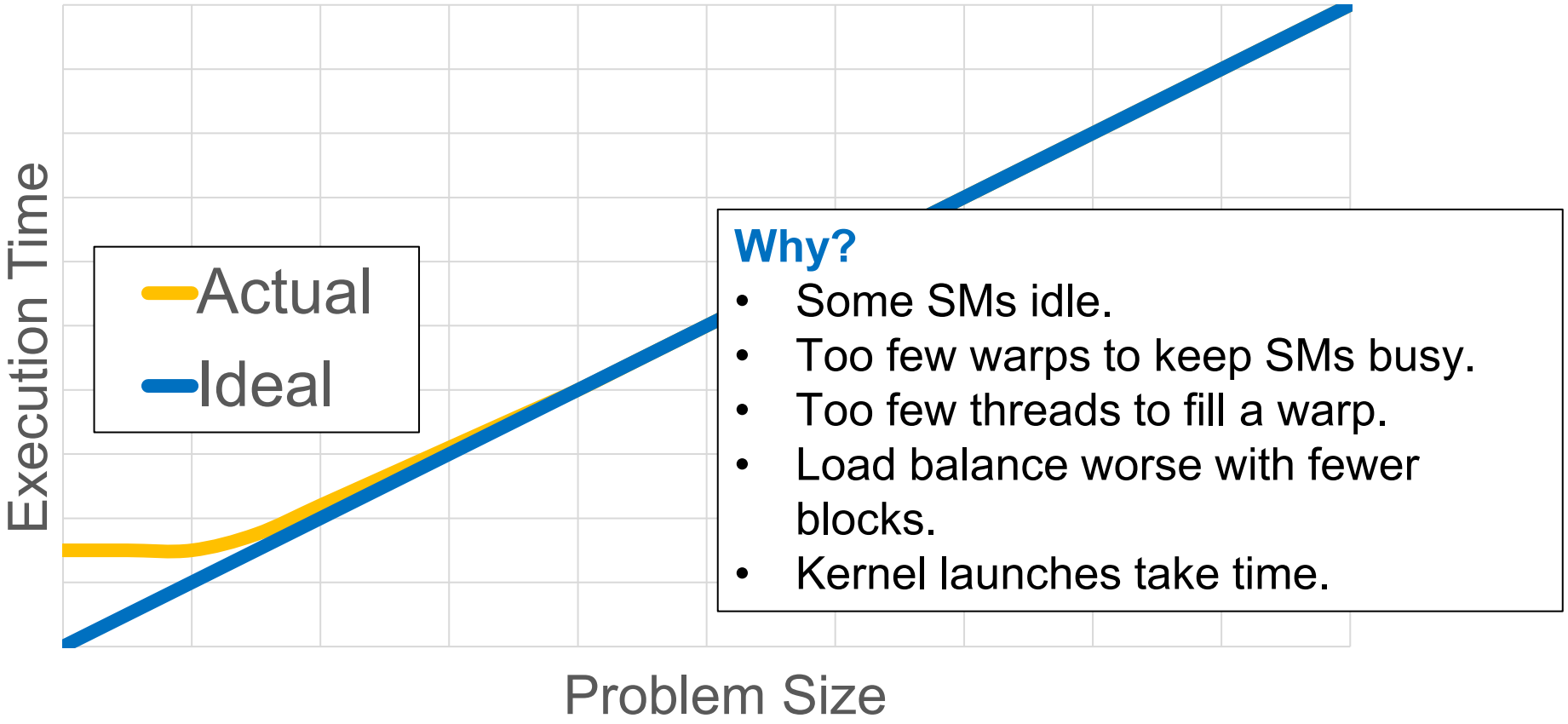


Why?

- Some SMs idle.
- Too few warps to keep SMs busy.
- Too few threads to fill a warp.
- Load balance worse with fewer blocks.
- Kernel launches take time.

Use Moderate Segment Size

- **Data transfers**
  - **have similar non-linearities** for small sizes
  - due to startup costs on host and DMA.

- **So how small should segments be?**
- **Moderately sized.**
- Best size likely to **depend on GPU**.

Any More Questions?

Things to Read:
CCPG: 3.2.8. Asynchronous Concurrent Execution

Things to Do:
Project CP2

Problem Solving

- You are tasked with performing some operations on a very long vector. Upon profiling your kernel code with nsys, you realize that a significant portion of the execution time is taken up by serial memory transfers to and from the device. You decide to use CUDA streams to overlap the memory transfers with computation to decrease the total execution time. There is a constant overhead for starting a kernel computation on the GPU, which is independent of the size of the data that the kernel is processing. (Note that kernel launch cannot overlap with kernel execution.) The following diagram illustrates the timeline for a single stream execution (not to scale).

Input Transfer A (Host to Device)	Kernel Launch Overhead	Compute Time	Output Transfer B (Device to Host)

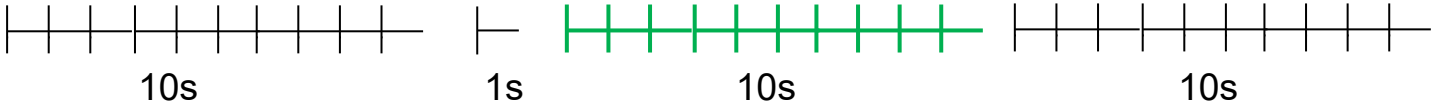
The times for each of the section shown above is as follows:

- Total Input Transfer time: 10 sec
- Constant Kernel Launch overhead: 1 sec
- Total Compute time: 10 sec
- Total Output Transfer Time: 10 sec

- Q: Assume that you have access to 3 hardware streams that enable continuous pipelining. How many segments should the input vector be divided into so as to minimize the total execution time where continuous pipelining is possible?

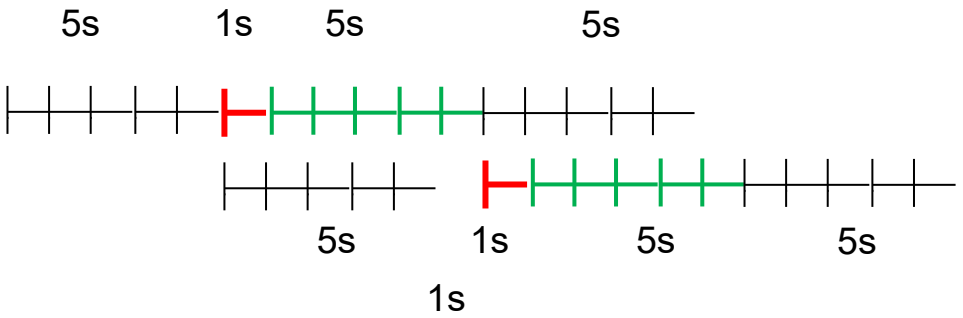
Problem Solving

1 segment
1 stream



31s

2 segments
2 streams



22s